

ccint — Canada Cyber Social Intelligence

A social-media cybersecurity sensor that measures its own coverage, and an agent layer that reports it

Full report

23 September 2026

In one paragraph. ccint collects Canada-related cybersecurity discussion from a configured set of sources, annotates it under coexisting versioned schemes, and tests whether anything changes. Its headline statistical result is negative and well characterised: no topic movement in the window is distinguishable from noise. This report is mostly about what was built to find out *why*, and what that machinery turned up. Measured against an external incident registry, the platform carries 26 of 27 Canadian ransomware events and the collector captured 26 of them — but only two drew a post from a non-automated account. What reads as coverage is syndication. A second, independent measurement points the other way: vulnerability identifiers that CISA has confirmed as exploited make up 2.7% of those mentioned and attract 19.3% of the mentions, so attention does track something real when the ground truth is not a feed. The final layer turns these components into ten tools an agent calls to answer questions in plain language, with the project’s measured constraints written into the system prompt so that the answers cannot quietly outrun the evidence.

1 The question, and why it changed shape

The project was framed as a longitudinal study: how does Canada-related cybersecurity discussion change over time on social media? A collector runs continuously, posts are labelled under versioned annotation schemes, and change is tested statistically.

The first full analysis returned a negative result, and returned it carefully. The largest topic movement in the 30-day window, `data_breach` at +22%, has $p_{\text{FWER}} = 0.734$ under an author-level circular-shift null that preserves each author’s volume, topic mix and burstiness while destroying calendar alignment. Empirical power analysis — measured by author-pool replication rather than assumed — put the requirement at roughly $8\times$ the current volume.

A negative result of that kind is only worth something if you know what produced it. Three explanations were open:

1. **The world.** Canadian cyber discussion genuinely has no structure at this timescale.
2. **The gate.** The relevance judge is too strict, so the corpus is a biased subsample of what was collected.
3. **The boundary.** The collector never retrieved the relevant material, so no judge could recover it.

Nothing in the system could separate them, because every number it produced came from labels it had produced itself. Most of what follows is the construction of measurements that can, and the answer was not the expected one.

2 Why the system is built this way

Four commitments shaped every design decision. Each exists to prevent a specific failure that would be invisible after the fact.

One platform, and say so. Bluesky was chosen for an open, documented API with usable historical search and terms that permit research collection; X and Reddit were excluded on access grounds rather than merit. The consequence is stated rather than hidden: this is a convenience sample, and a large part of the project’s job became measuring what such a sample can and cannot show. Two syndicated feeds were added later, which is where the multi-source architecture came from.

Labels are versioned and never overwritten. Annotation rows are keyed by (post, annotator version) and only ever inserted; eight annotator versions now coexist over the same corpus and diff row by row. Without this, every improvement to the annotator would silently rewrite history, and a shift in topic distribution could never be attributed to the world rather than to the method.

Raw data is verbatim; every number carries provenance. Payloads are stored unmodified, and every run and model label records what produced it — including the SHA-256 of the exact prompt text. That last part was added after noticing that a model annotator’s output is entirely determined by its prompt: “labels are never overwritten” had preserved the labels while losing the conditions that made them.

System failure must be distinguishable from social change. Every collection path records mode, status, window and counts, so a day with half the usual volume is attributable to either the world or a token expiry. The principle held under test: a token outage appears in the record as 43 partial runs, not as a drop in Canadian cyber discussion.

Underlying all four: **measure before improving.** Repeatedly the intuitive next step turned out to be aimed at the wrong quantity, and only measurement revealed it. Tuning the trend score would have been wasted effort; relaxing the relevance rules helps, but not for the population that appeared to need it; and the single largest prompt improvement in the project introduced a regression that the external evaluation set certified as correct.

3 What was built

Collection runs in two modes against the term list — a day-by-day backfill and an hourly incremental — with idempotent ingestion, stale-run reaping and client-timestamp repair. The corpus stands at 96,002 posts across three sources. Annotation proceeded in generations, all retained: two lexicon-based, one that re-decides topic only while inheriting the relevance verdict verbatim so that the two changes stay separately attributable, and one that replaces the relevance decision entirely. Analysis covers descriptive statistics, permutation testing against two null models, behavioural broadcast detection with a collection-lag control, and a two-axis actor typology that separates measured audience from content-judged function.

Sources are configuration, not code. `config/sources.yaml` names the forums and feeds to read; two generic collectors — RSS/Atom and the Discourse public JSON API — cover most of what a security community exposes. Both consult `robots.txt` and refuse a source they cannot read, because a collector that gets blocked produces a gap, and a gap is indistinguishable from a fall in discussion, which is the confusion this project exists to prevent. Each source must declare why automated access is believed permissible; the loader rejects one that does not. Web sources reuse

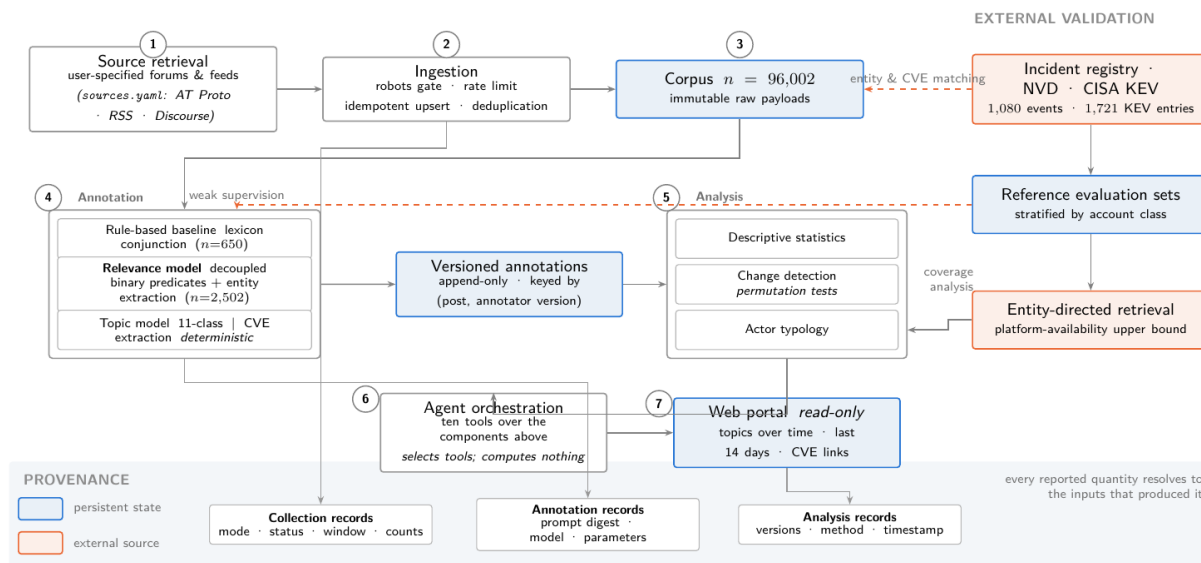


Figure 1: Processing pipeline. Stages 1–3 acquire and store the corpus, stage 4 annotates it under coexisting versioned schemes, stage 5 analyses the result, and stages 6–7 expose it. The external-validation column supplies labels no component of the pipeline produced; the provenance band records what produced every reported quantity.

the existing ingest path rather than adding a second, since a second path is a second way to bypass run bookkeeping.

4 The collection boundary had never been examined

The collector issues one search per term over a fixed list of 34 cybersecurity terms — 25 English, 9 French — and merges the results. None of the 34 mentions Canada. The list had not changed since the first day of the project; the `query_version` mechanism faithfully recorded that it was `cyber_v1`, which guarantees reproducibility but says nothing about adequacy.

Because platform search is close to text matching, per-term productivity can be reconstructed offline from the stored corpus without new collection. Figure 2 plots what each term cost against what it returned.

Three properties matter more than the ranking:

- CVE retrieved 12,502 posts and yielded 3 relevant ones, a precision of 0.024%. It accounts for 13.7% of everything collected.
- Eleven terms returned zero relevant posts. One of them, `maliciel`, returned no posts at all in 30 days.
- `cybersecurity` alone accounts for 30.8% of all relevant posts and 24.6% of the entire corpus. The time series of the corpus is, to first order, the time series of one search term. Nothing monitored it.

The instructive part is the provenance of the list. The project’s specification named four term pairs and wrote “etc.”; the remaining 26 terms were supplied during implementation from general domain knowledge. They were never derived from data, never compared against

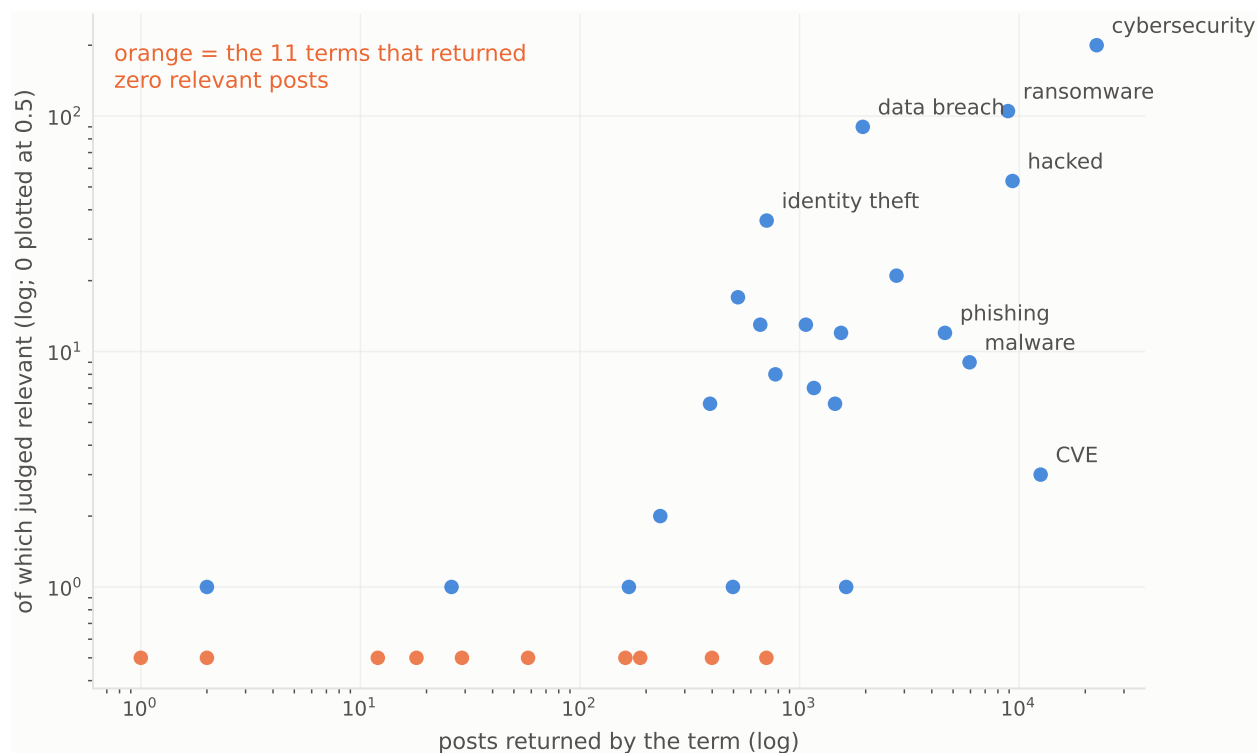


Figure 2: Each search term: posts retrieved against posts subsequently judged relevant. Both axes are logarithmic; terms with zero relevant posts are drawn at 0.5 so they remain visible. The distribution is not merely uneven — it is close to degenerate.

an external list, and never revised. The published work that does this well does one of two things: it inherits a list from practitioners — Tweezers (NDSS 2025) takes its keywords from a threat intelligence company’s analysts — or it grows one from a seed by measured expansion. Tweezers also reports what such lists cost: comparing its own against the earlier W2E list, more than half of the tweets it collected would have been missed by W2E, and W2E was itself missing *phishing*, *cve* and *patch*. Two lists built by professionals disagreed by more than half.

5 The relevance gate, replaced and then measured

The old gate was `is_relevant = has_cyber_term AND has_canada_term AND corroborated`. Both sides are narrow instruments joined by a conjunction, so recall is bounded by their intersection: of 91,551 posts collected at the time of the measurement, 650 passed (0.72%), and the 89,714 rejected had never been examined by anything semantic.

The replacement asks a model two questions, and the first design decision was to keep them apart. A single prompt asking whether a post is about *cybersecurity in Canada* produced a 6.8% pass rate on previously rejected posts, and inspection showed why: the model had collapsed the conjunction onto the side it knows better. The passing posts were vendor advisories, ransomware victim listings, and a French story about the French Ministry of Education. Asked as two independent questions, the Canadian side passed 1.08%, and the passing posts began to include material that should have been there all along — a Quebec cabinet minister responsible for cybersecurity, *.ca* law-firm domains, “GC website” as colloquial shorthand for the federal government.

A second decision follows from the corpus rather than from the model: ask about Canada first,

over everything, and ask about cyber only where a Canadian entity was found. Every post in this corpus arrived through a cybersecurity search term, so the cyber side is nearly always true and carries little information; the Canadian side is the rare and binding one. Running it this way also produces an entity map of the whole corpus as a by-product — the input to query expansion and to event linking — and it reduces the call count from $2N$ to $N + \varepsilon N$.

5.1 The gate’s own failure

The first version of the entity prompt recovered Canadian entities in only 55.4% of registry-verified positive posts and — more diagnostically — in only 69.0% of posts containing the literal string *Canada* or *Canadian*. Specificity was excellent at 99.5%. The failure was therefore recall, and its cause was the prompt: a clause instructing the model that an empty list is the expected answer, followed by a long list of prohibitions, drove a 4B model to the null answer. Removing the restraint clause and adding an explicit rule — if the country name appears literally, it counts — moved literal-mention recall from 69.0% to 99.3% (Figure 3).

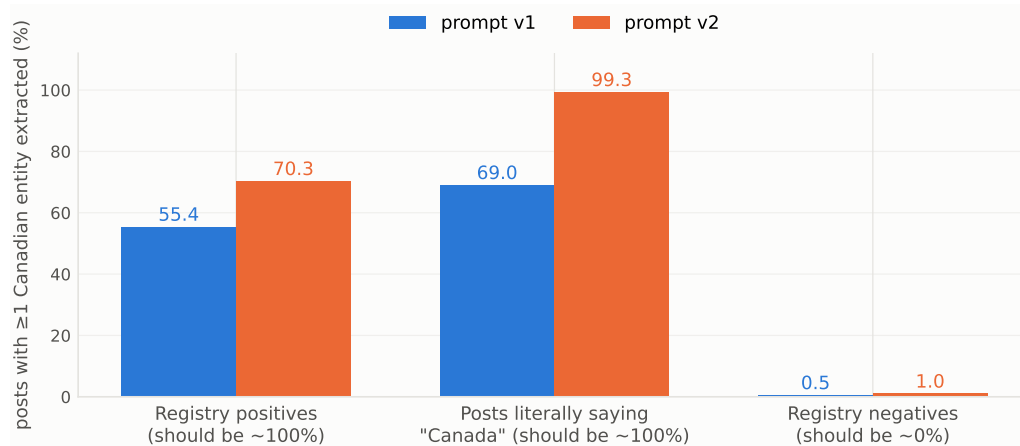


Figure 3: Entity recall before and after the prompt revision, on posts that contain the country name literally. The revision fixed the failure it targeted, and introduced a second one that no evaluation set in the project could see.

5.2 And the fix broke something the evaluation could not see

The corrected prompt raised corpus relevance from 906 posts to 3,528. That number is wrong. Inspection of the additions found *trustedsec.com*, *malwarebytes.com*, the UK’s NCSC, *Revolut*, a French story about the *Éducation nationale*, and *ia.cr* — the Costa Rican top-level domain. The prompt’s clause “*a .ca or .gc.ca domain*” had been generalised by the model into “*a domain*”.

Two mechanical checks, requiring no further model calls, quantify it:

Extracted entity type	Count	Not present in the post text
place	2,192	1,053 (48.0%)
org	1,364	12 (0.9%)
domain	1,296	49 (3.8%)
gov	889	10 (1.1%)

Of the 1,296 **domain** entities, 312 (24.1%) actually end in a Canadian TLD.

A verification channel obtained by accident. Character offsets are computed with `str.find` rather than requested from the model, on the narrow grounds that small models miscount offsets. That decision turned into an independent hallucination detector: an entity string that cannot be located in the post was invented. Nearly half of all `place` entities were — the model writing “Canada” into a French post about a French ministry. Declining to trust a model with a task it does badly produced a free check on the task it was actually given.

Applying both rules — discard entities absent from the text, discard domains outside Canadian TLDs — removes 2,086 entities and 1,180 posts. The validation layer is retained in the labeler even though the prompt has been corrected, because it is deterministic, costs nothing, and catches exactly the class of error the external evaluation set cannot.

Relevance gate	recall @ registry pos.	specificity @ registry neg.	corpus relevant
rules_v2 — keyword conjunction	46.5%	99.8%	650
llm_v2a — two questions, prompt v1	53.5%	99.7%	860
llm_v2c — prompt v2 + validation	68.3%	99.4%	2,348

before validation, prompt v2 reported 69.3% / 99.0% / 3,528 — one third of that corpus was spurious.

The negative set certified a corpus that was a third wrong. A specificity of 99.0% on the registry negative set implies roughly 890 false positives across 89,044 posts. The true figure was 1,181. The negative set did not detect the failure because it is composed of feed posts about foreign ransomware victims: formulaic English, few URLs, no French. The model’s actual failure modes — over-generalising a domain rule, hallucinating a country name into French text — do not occur in that population. An external label removes self-reference; it does not confer representativeness, and a non-representative evaluation set does not merely fail to detect a regression, it actively certifies one.

5.3 What remains is a knowledge ceiling

Of the posts the corrected and validated extractor still misses, **none** contains the word *Canada* in any form. They read “*New post from #Qilin: Coldfish Seafood*” — a real Canadian victim, named, with no country marker anywhere in the text. No amount of prompting will let a 4B model know that Coldfish Seafood is Canadian.

This is a ceiling of a particular kind. The residual error in the relevance gate is not a model-capability problem and not a prompt problem: it is missing world knowledge about small Canadian organisations. The only thing that resolves it is a gazetteer — and the gazetteer that would resolve it is the external registry currently being used to measure the gap. The measurement instrument and the repair are the same object.

6 An external reference, and what it revealed about itself

Until this phase, every precision and recall figure in the project traced back to a single author: the person who wrote the rules also wrote the prompts and produced the hand-coded evaluation set. A

reported $\kappa = 0.870$ measures agreement under those conditions, not independence.

An incident registry breaks that loop. Victim records were ingested for Canada (positives) and the United States, France and the United Kingdom (negative entity sources): 1,080 events and 1,943 entities. Matching victim organisation names against the corpus produces evaluation sets whose labels no part of this system produced.

Two constraints were imposed on their construction, and both mattered more than expected.

First, the definition of an evaluation set may not contain the output of the system being evaluated. A natural specification for the negative set was “mentions a foreign victim, and `rules_v2` judged `is_canada` false”. That specification makes `rules_v2` precision identically 1 on the set, and any replacement can only appear worse — an artefact of construction with no relation to capability. The negative set uses registry conditions only.

Second, the positive set must be stratified by account type. The reason is visible in the data: registry victim names appear on the platform almost exclusively inside the accounts that syndicate the registry. Of 101 positive posts, 100 come from automated feeds. Median engagement is zero and 94.9% of the posts have no likes at all. The single post classified as non-feed turned out to be a golf-industry content farm auto-generating an article about a golf club’s ransomware incident, with zero likes, reposts and replies.

A metric that rewards capturing bots. Had recall on the registry positive set been adopted as the optimisation target, it would have climbed towards 100%, because those posts are formulaic and a model need only learn which small firms are Canadian. The metric would improve substantially while the research question — what Canadians say about cyber incidents — received no answer at all. External labels are necessary but not sufficient: an external label attached to a non-independent *population* reintroduces the circularity it was meant to remove.

7 The central result

Combining the registry with a targeted per-event search over the platform separates the three explanations. For every Canadian event in the window, the victim organisation’s name was searched directly — with no cybersecurity term attached — over a $[-2, +30]$ day window. Because bare organisation names attract unrelated traffic (*Air Canada* retrieves fare deals), every hit was then passed through the cyber judgement before counting. This is entity-directed retrieval: it establishes an upper bound on what the platform could have yielded for an event whose identity is already known.

Stage	Events	Share
Listed in the registry, in window	27	—
At least one cyber-related post exists on the platform	26	96%
At least one such post was collected by ccint	26	96%
At least one such post came from a non-feed account	2	7%

The 144 cyber-related posts across these events come from 15 accounts. Twelve are automated threat-intelligence feeds. The three that are not are Canadian news organisations, contributing five posts between them, all about the same event. Individual users contributed nothing.

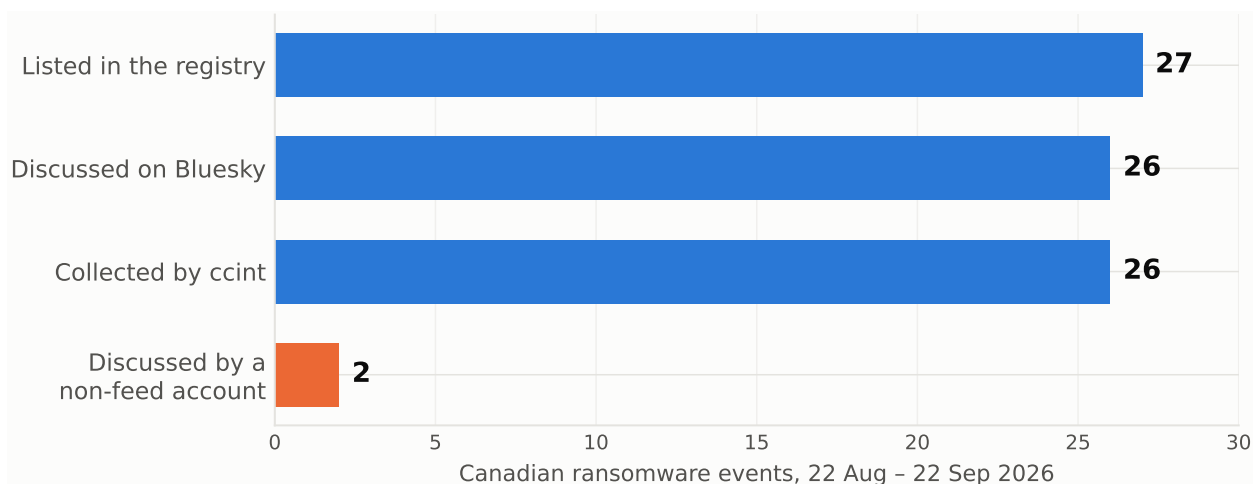


Figure 4: The same 27 Canadian ransomware events at four successive gates. The attrition is almost entirely at the last one, and the last one is not a property of this system.

The collector is not the bottleneck for this event class. It captured 26 of the 26 events the platform carried. The reason the corpus contains no detectable trend in ransomware discussion is that there is almost no ransomware *discussion* to detect — there is a nearly complete mirror of several leak-site feeds, and the corpus reproduces it faithfully. That also reframes the power analysis: $8\times$ of this corpus is $8\times$ as much syndication, so volume is the wrong lever.

7.1 The one event that behaved differently

The exception is worth stating in full, because it is the only case in the month where the platform behaved the way the project assumed it would.

The Commission de la construction du Québec, a provincial construction industry regulator, was listed by the Qilin ransomware group; roughly 350,000 people’s data was taken. This event drew genuine Canadian media coverage in French and English: a class action filed in Quebec Superior Court, reporting on the scale of the theft, and a radio interview with the CCQ’s chief executive.

Four of those five posts are in the corpus. The one that is not reads “*Une protection préventive mise en place à la CCQ: «Je me sens responsable »*”. It contains no term from the 34-term list — *protection préventive* is in no lexicon — and so it was unreachable. This is the collection-boundary gap at human scale: one post, one missing phrase, a real event, a real Canadian outlet.

7.2 Two gaps, and a reference that can only see one

The measurements point at two distinct deficiencies, and conflating them would be a mistake.

	Registry-verifiable events	General Canadian cyber talk
Collection recall	26/27 events (96%)	unmeasured; at least 901 rejected posts carry a Canadian entity
Binding constraint	the platform has no organic discussion	the Canada lexicon misses entities
Fixable by better queries?	no	yes
Visible to the registry?	yes	no

A full-corpus screen of the 83,370 rejected posts long enough to judge found 901 containing a Canadian entity; attribution shows 55.8% failed because `is_canada` was false — the lexicon missed the entity — and *none* were blocked by the corroboration rule that a prior review had proposed retiring. Of the 1,799 posts the new gate newly admits, 84.0% were rejected by the Canadian side of the conjunction, against 12.2% on the cyber side and 3.7% on both; in the other direction the new gate rejects 138 posts the rules had admitted. The asymmetry is the point: nearly all recoverable recall sat behind the Canada lexicon.

But the registry cannot see that population at all. It contains ransomware victims, so it can only score the class of discussion that ransomware feeds dominate.

8 Linking discussion to vulnerabilities

CVE identifiers are extracted by regular expression rather than by a model. Their format is fixed by MITRE, so extraction is deterministic; this is the one table in the system whose trustworthiness is independent of annotation quality, which is unusual here and worth stating plainly. Identifiers are then enriched from two external sources the system cannot influence: NVD for theoretical severity, and the CISA Known Exploited Vulnerabilities catalogue for the fact of observed exploitation. The two are kept in separate columns and never combined into a risk score. One is a judgement about how bad a flaw could be; the other is an observation that somebody is using it. A combined number would hide which half is external fact — the same error, in a smaller form, as the composite trend score this project rejected earlier.

Across the corpus, 22 Aug – 23 Sep 2026	Count	Share
Posts mentioning at least one CVE identifier	10,820	—
Distinct identifiers mentioned	5,404	—
Of those, listed in the CISA KEV catalogue	147	2.7%
Share of all 12,055 mentions going to those 147	—	19.3%
Of the 25 most-discussed identifiers, in KEV	21	84%

Two caveats belong with that figure. The concentration is measured over mentions, and the feed accounts characterised above contribute a large share of them, so this is partly a statement about what threat-intelligence syndication prioritises. And the two enrichments are not equally complete: one catalogue fetch covers KEV status for every identifier, whereas NVD severity is subject to a rate limit and is currently resolved for 379 of the 5,404 mentioned identifiers, prioritised by corpus prominence rather than by ease of retrieval.

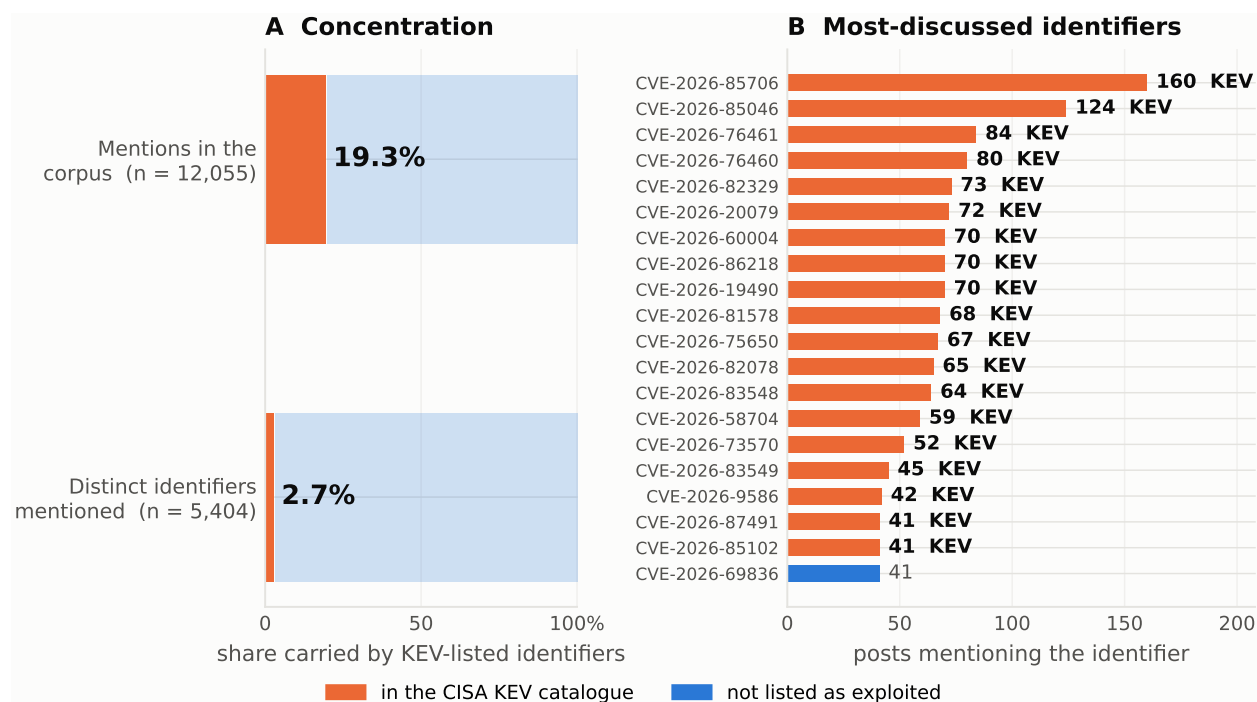


Figure 5: Attention against confirmed exploitation. **A:** identifiers in the KEV catalogue are 2.7% of those mentioned but carry 19.3% of the mentions. **B:** of the twenty most-discussed identifiers, nineteen are KEV-listed. The two panels are deliberately separate charts — one is a proportion and the other a count, and a single axis carrying both would mislead.

Attention tracks exploitation rather than severity. A set of vulnerabilities making up 2.7% of those mentioned attracts 19.3% of the mentions — a concentration of roughly sevenfold — and more than four in five of the most-discussed identifiers are ones CISA has independently confirmed as exploited. This is the first result in the project where social-media attention aligns with an external ground truth instead of mirroring a feed, and the first that is directly usable: ranking vulnerabilities by how much they are discussed recovers most of the confirmed-exploited set without being told what it is.

9 Making the system answerable

The measurements above are only useful if somebody other than their author can interrogate them. The final layer adds no analysis. It turns the existing components into ten tools, gives a model the ability to call them, and puts the results behind a read-only portal.

The tools compute nothing of their own. `get_actor_composition` forwards to the same broadcast detector the analysis uses, with the same thresholds; `collect_sources` forwards to `run_collection`, so an agent-triggered collection lands in the run record like any other and cannot become an untracked gap; all SQL lives in one query module. This is enforced by test rather than by convention: the suite fails if a second copy of the broadcast threshold appears in the tool layer, because two implementations of one definition drift apart and nobody notices until a number is wrong in public.

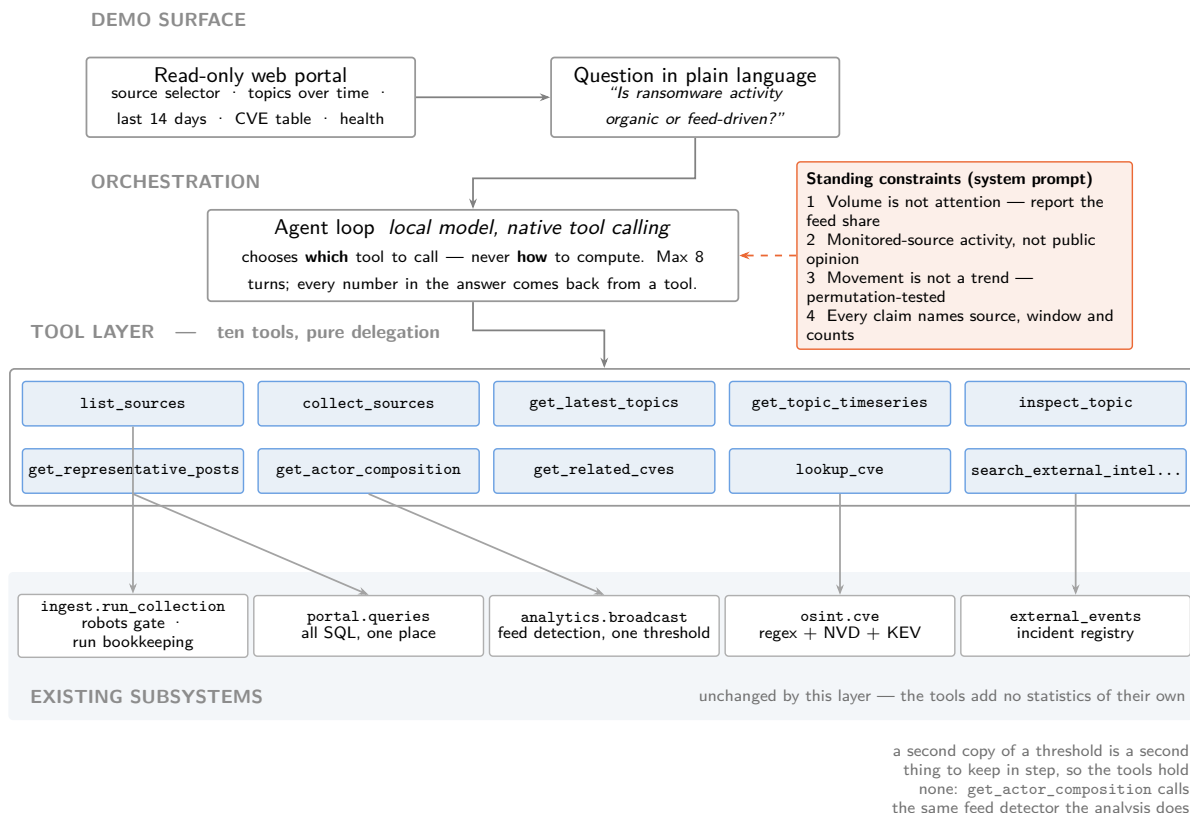


Figure 6: The orchestration layer. The agent chooses *which* tool to call; it never chooses *how* to compute. Every statistic is produced by the same subsystem that produces it for the analysis pipeline, so a question asked twice cannot be answered on two different definitions.

The agent’s constraints are the project’s measurements. Four findings are written into the system prompt as standing constraints: volume is not attention and the feed share must be reported alongside it; this corpus describes activity in the monitored sources and not Canadian public opinion; permutation testing found no movement distinguishable from noise, so an increase is not a trend; and every claim must name its source, window and counts. Severity and exploitation must be reported separately. If the backend is unavailable the agent refuses rather than answering from memory.

Asked “*Is ransomware activity mostly organic or feed-driven?*”, the agent called `list_sources`, `get_latest_topics` and `inspect_topic` across four turns and answered that activity is mostly feed-driven, quoting the measured broadcast share, limiting the claim to the monitored sources, and declining to describe the window-over-window change as a trend. The interesting part is not that the model can call tools; it is that the constraint set makes the honest answer the only one the tools support.

What the portal shows. A source selector filters every view, so a demonstration does not require editing configuration. Beside the content it shows collection health, because a quiet day and a broken collector look identical on a topic chart and different in the run record. Every panel carries its own caveat text, since a bare number on a screen reads as unconditional. Collection can be triggered only for sources already declared in configuration; an arbitrary URL is rejected.

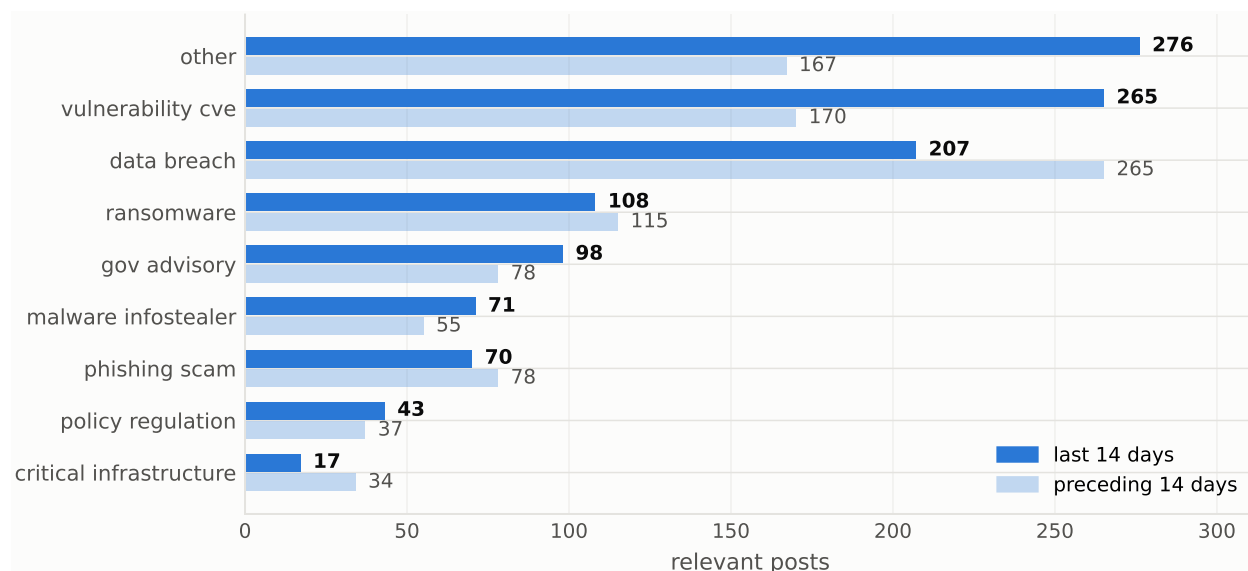


Figure 7: The last fourteen days against the preceding fourteen, as the portal presents them. These are descriptive counts. Permutation testing on this corpus found no topic change distinguishable from noise, so no bar here should be read as a trend — which is exactly why the portal labels them as activity rather than as change.

Building this layer surfaced two defects in the analysis it wraps, both of the kind that only appear when numbers are put on a screen for someone else. Topic queries folded posts carrying no topic row into the residual *other* bucket — the same error the project had already ruled out in its actor typology, where *unclassified* must never be read as a negative. Labelling them (*unlabelled*) immediately exposed 962 posts in the recent window with no topic label at all, which were then labelled. And the portal’s topic version constant still pointed at a superseded annotator, so it had been reading a scheme covering a fraction of the corpus.

10 What the project is

The original framing makes a weaker claim available and a stronger one unavailable.

Unavailable: trend analysis of Canadian cyber discourse on this platform, at this volume, for this event class. Not because the statistics are under-powered — though they are — but because the underlying signal is mostly machine output, and adding data would add feed posts.

Available, and apparently unusual: a sensor that reports its own blind spots with numbers. Published systems in this area report how many events they detected — 84 of 151 in the strongest comparable work, 45 and 4 in the systems it compares against. All three report a numerator. None reports how many events had no platform presence at all, because establishing that requires the per-event entity-directed retrieval run here, and that is only affordable when the event count is small. This project’s greatest liability — a corpus of a few thousand relevant posts — is exactly what makes the denominator measurable.

The defensible contribution is not “we detected Canadian cyber events on social media”. It is “here is what a single-platform social sensor can and cannot see about a national cybersecurity picture, measured against external references, with the syndication component separated from the organic one, and exposed through an interface that will not overstate it”. The first claim is

weak and partly false. The second is, as far as the surveyed literature goes, unreported.

11 Limitations

- **One registry, one event class.** The incident registry lists only victims named on leak sites. Breaches disclosed by regulators, phishing campaigns, advisories and policy events are invisible to it, as are victims who paid. Every figure conditioned on the registry describes ransomware coverage, not Canadian cybersecurity coverage.
- $N = 27$. The three-stage decomposition rests on 27 events in one 30-day window. The qualitative pattern is stark, but the percentages carry wide intervals and should not be quoted as point estimates.
- **Feed classification is a heuristic.** Accounts are labelled automated by volume and near-zero engagement. The golf content farm shows the failure mode: a low-volume automated account scores as organic, which biases the organic count *upward*. The true organic count is therefore no higher than reported, and probably lower.
- **No independent human annotation yet.** A 200-post stratified audit sample has been drawn and deliberately left unlabelled. The person who wrote the prompts should not be the person who scores them; doing so would reproduce the circularity the registry was introduced to break.
- **The evaluation sets are narrow in a way that matters.** Both are built from ransomware victim names, so they exercise a formulaic English subpopulation. The domain over-generalisation documented above passed them cleanly. Any future prompt change should also be checked against a probe drawn from the corpus itself.
- **Multi-source collection is implemented but not demonstrated at scale.** The two feeds added so far contributed 45 posts, because RSS exposes only recent items and cannot be backfilled. A comparable time series requires running them forward for weeks or adding a paginated source.
- **A 4B model.** Both gates and the agent run on a local Qwen3-4B. The residual gate failure — not knowing which small firms are Canadian — would partly yield to a larger model, though a gazetteer addresses it more directly and more cheaply. The agent's tool selection is reliable on the demonstration queries but has not been evaluated systematically.

12 What follows

1. **Widen the registry before widening anything else.** Advisories from the Canadian Centre for Cyber Security, KEV and Canadian media would extend the external reference past ransomware. This is the single change that would most alter the picture above; the current denominator is narrow in a way that flatters nothing.
2. **Feed the registry back into the gate.** The irreducible residual is missing knowledge of Canadian organisation names, and the registry is a list of exactly those names. This closes a loop currently open in the wrong direction.
3. **Add a Canada-side query stream** as an additional disjunctive stream rather than a filter — not to improve ransomware coverage, which is already at 96%, but to reach the population the registry cannot see: the CCQ interview that no cybersecurity term could retrieve.

4. **Have someone else label the audit sample.** It is the only remaining step that cannot be automated, and the one every other number depends on.
5. **Let the new sources accumulate,** and add a paginated one so that multi-source comparison rests on more than 45 posts.
6. **Report absolute counts, never recall ratios.** With a denominator this small and this biased, a ratio invites a comparison the data cannot support.

Provenance. The corpus stands at 96,002 posts across three sources, 22 August – 23 September 2026. Gate evaluation figures were measured on 22 September against a corpus of 91,551 posts and are left at the values recorded then; the incident registry and the KEV catalogue were retrieved 22–23 September. Eight annotator versions coexist over the corpus; every model-produced label carries the SHA-256 of the prompt that produced it. The test suite stands at 268 passing. Evaluation sets are distributed as post identifiers with labels, not as post text.